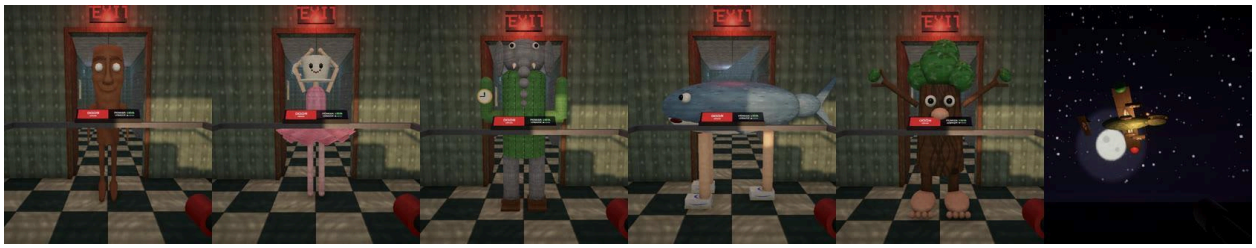
ELEC 328 Project Report
Assn 2 - Enchanted VR Flashlight
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1. Concept



[Header Image(s) (can reuse a screenshot below if you'd like)

Title: Five Nights At Brainrot

In one short paragraph (3-5 sentences), describe the learning objectives of the assignment and the project you created to satisfy it. Write this in your own words.

This assignment was about lightning and shaders: building an enchanted flashlight in VR with a beam that reveals a hidden world. The main goals were connecting a C# script to a shader so that the shader knows where the flashlight is in every frame using the tools and equations that we learned in class. To make it more fun, I turned it into a game called Five Nights at Brainrot. The game is similar to Five Nights at Freddy's. You worked the night shift in an office and the goal is to survive from 12 AM to 6 AM while Triple T and five other Italian Brainrot characters try to catch you. The flashlight allows you to discover hidden cursed messages and also defend yourself, and each character has a different reaction to the light.

2. Technical overview

Explain the main system you built and the techniques that make it work. Use the prompts below that apply to your assignment.

[Write your technical overview here. Include only the techniques relevant to your assignment.]

Here I detail the techniques used for each part of the game and how they align with the assignment requirements

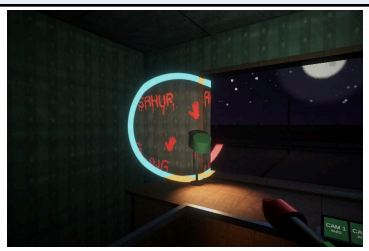
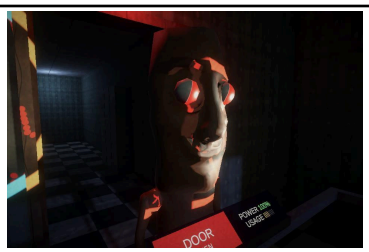
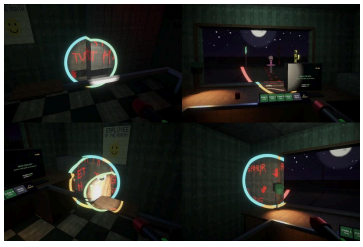
1. **Flashlight -> Shader (Objective A):** The flashlight is a grabbable object (XR Grababble). Its script sends the beam's worldspace origin, direction and cone angle and range to the shaders in Update every frame by changing the values of the Shader object. These values adapt continuously as global variables each time the scene updates.
2. **The Light Cone (Objectives B,C,E):** The Shader Graph has the properties Base Map, Hidden Texture, Boundary Color, Boundary Width and Edge Feather. A custom function node takes each pixel's world

position, makes the unit vector from the flashlight to that point, and takes its dot product with the beam direction.

3. **Surfaces facing away (Objective D):** The shader also identifies at which surface is facing. If it faces away from the flashlight, like the back of a character, it keeps its normal texture even when it is inside the beam
4. **Many materials and creative extras (Objective F):** There is a great number of materials that use the same shader, each with its own texture, hidden texture and outline color. For example, the wallpaper hides "TUNG TUNG TUNG" written in blood, the floor hides footprints and every character has its own hidden look.

3. Evidence of requirements

[Use this table to point to the exact evidence for each requirement. Include video timestamps and screenshot labels.]

Requirement	Where it is demonstrated	Screenshot
A: Grababble Flashlight	[Experience video 0:30 ; Screenshot 1]	
B,C,E: Default texture outside, hidden texture inside, solid color-outline	[Experience video 0:28; Screenshot 2]	
D: Surfaces facing away keep the default texture	[Experience video 1:06; Screenshot 1]	
F: multiple materials with different base map, hidden texture, boundary color	Screenshot (Fixed after recording video)	

5. Assets, data, and citations

List every external asset, dataset, model, texture, sound, library, or reference used. State whether it was created by you, provided by the course, or obtained elsewhere. Provide a source link or formal citation where appropriate.

Item	Source / citation	How it was used
Tung Tung Tung (Triple T) Sahur Asset	https://sketchfab.com/3d-models/tung-tung-tung-sahur-91ddd9079bd84019ba4a12e01d93a0d6	The enemy in the game is Triple T, and one has to shoot a laser to get rid of them. Triple T also dances and can do back flips.
Five other characters, all textures, the night sky, the sounds and the office	Created for this project (generated in code, see AI disclosure)	Characters were build with the help of Claude by merging simple shapes and textures such that it resembles the original characters
Homework 1 Scripts	Previous assignment	Tung's celebration dance and the 6-7 move
Five Night's at Freddy's	Inspiration from Five Nights at Freddy's by Scott Cawthon (2014)	Inspiration for the game design and the characters

6. AI-use disclosure

If you used generative AI during project development, disclose it below. If you did not use AI, write: No generative AI was used in project development.

Tool	Used for	What was incorporated, changed, or rejected
Claude	Programming the game: C# scripts, the Shader Graphs, scene builders and character builders, and sounds.	I came up with the concept of the game and made design decisions. Most of the new code and character designs were made by claude, and I incorporated as jumpscares the animation I had from the previous assignment for Triple T. Claude helped generate the scene and the designs as well.

7. Reflection

[Briefly explain what worked, what was difficult or limited, and what you would improve or explore next. Write what you learned. It's fine to be repetitive with the Concept/Technical overview sections. Write this reflection in your own words.]

What worked the best was the effect of the light for revealing and flashing characters. Once the flashlights sends its position and direction every frame, the code just had to do a dot product and angle comparison, and it was very

nice to be able to sweep the light around the office and see how each character had a different hidden message when you flash at them. The hardest part of this project was understanding what the C# script and the Shader graph were doing after Claude wrote them, and also understanding how these two programs talked to each other. It was very interesting to see how the math that we learned in class would show up in a VR experience.

One limitation is that the new characters are built from simple shapes instead of being dedicated models made in blender or some other program. Maybe for the next versions, I will invest a bit of money into buying a dedicated model and making the game look much nicer. My main takeaway from this assignment is a better understanding of shaders and light and how one needs both the Shader Graph and a C# script talking to each other to make everything work well.